SQL START!

ANCONA, 29 Settembre 2012

SQL Server 2012 AlwaysOn FCI & AG

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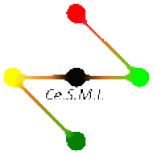
Microsoft SQL Server

Sponsors

Microsoft



Speaker

▶ Gianluca Hotz

- Consultant and mentor at SolidQ
  - Administration, capacity planning, data modeling, development, optimization
  - >15 years on SQL Server (from 4.21 in 1996)
- Interests
  - Relational model, DBMS architectures, high availability and disaster recovery
- Microsoft MVP SQL Server since 1998
- Founder & vice-president UGISS
  - User Group Italiano SQL Server
- Mail: ghotz@ugiss.org - ghotz@solidq.com

Microsoft SQL Server

AlwaysOn

▶ Introduction

- New name for HA and DR technologies
  - Failover Cluster Instance (FCI)
  - Availability Groups (AG)
- Previously named HADRON
  - Keep in mind when searching articles
  - Still present in some DMVs (_hadr_)
- For the purpose of this presentation
  - High Availability (local HA): in a data center
  - Disaster Recovery: across data centers

Microsoft SQL Server

AlwaysOn FCI

▶ Brief review

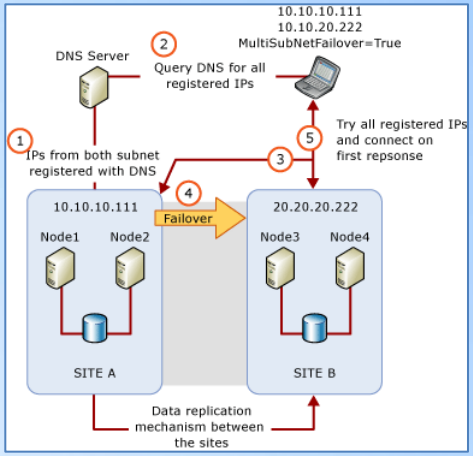
- Protects an entire SQL Server instance
- Requires Shared Storage
- Geographic DR with SQL Server 2008
  - Proprietary solutions for Stretch V-LAN
  - Proprietary solutions for storage replication

Microsoft SQL Server

AlwaysOn FCI

▶ Multi-Subnet Support

- Nodes on different subnets
  - IP Address resources in OR
  - Automatic DNS updates
- How do clients reconnect?
  - New connection string parameter
    - MultiSubnetFailover=True
  - For old clients
    - Reduce timeout value
    - Reduce DNS updates latency (HostRecordTTL)
    - Reduce DNS cache expiration on the client
- Still need storage replication technology



http://msdn.microsoft.com/en-us/library/ff878716.aspx

Microsoft SQL Server

AlwaysOn FCI

▶ Failover policies in SQL Server 2008

- Looks Alive
  - Checks service state (Service Control Mgr.)
  - Every 5 seconds
- IsAlive
  - Checks SELECT @@SERVERNAME
  - Every 60 seconds
  - Can pass check even when not responding to other incoming requests
    - e.g. stuck schedulers

Microsoft SQL Server

AlwaysOn FCI

▶ Flexible failover policies in SQL Server 2012

- Resource DLL calls system procedure
  - sp_server_diagnostic
- Looks Alive, IsAlive based on
  - Diagnostic information from procedure
    - New instance parameter FailureConditionLevel
    - Allows selecting conditions for failover to occur
  - Timeouts while calling diagnostic procedure
    - New Cluster parameter HealthCheckTimeout

Microsoft SQL Server

AlwaysOn FCI

▶ Failure condition levels

Level	Condition	Description
0	No automatic failover or restart	Never Fail, Log Only
1	Failover or restart on SQL Server down	Server Down
2	Failover or restart on SQL Server unresponsive	Server Hang (diagnostics sp not responding within HealthCheckTimeout)
3	Failover or restart on critical SQL Server errors	System Unhealthy (sp returns system error)
4	Failover or restart on moderate SQL Server errors	Resource Unhealthy (sp returns resource error)
5	Failover or restart on any qualified failure	QP Unhealthy (sp returns query_proc error)

Microsoft SQL Server

AlwaysOn FCI

▶ Flexible failover policies configuration

- Failover Cluster Manager
- PowerShell
- T-SQL!
  - ALTER SERVER CONFIGURATION FAILOVER CLUSTER PROPERTY HEALTHCHECKTIMEOUT = 60000;
  - ALTER SERVER CONFIGURATION SET FAILOVER CLUSTER PROPERTY FAILURECONDITIONLEVEL = 3;

Microsoft SQL Server

AlwaysOn FCI

▶ Reducing planned downtime with SQL Server 2012

- Windows Core Support
  - Less patch to be installed (circa 50-60%)
- Support for rolling upgrade and patching
- Better control on failover time
  - Indirect Checkpoints
  - ALTER DATABASE …
    SET TARGET_RECOVERY_TIME = …

Microsoft SQL Server

AlwaysOn FCI

▶ New Storage options with SQL Server 2012

- Tempdb on local storage
  - Can leverage fast SSD cards
- SMB shares supported
  - Best on
    - Windows Server 2008 R2 (SMB 2.1 + Hotfix)
    - Windows Server 2012
- Asymmetric Disks

AlwaysOn AG

▶ Data replicas review

- Log Shipping
- Database Mirroring
- Replication
- SQL Server 2012: Availability Groups

Microsoft SQL Server

AlwaysOn AG

▶ Common Problems

- Manual Failover*
- No coordination to failover multiple DBs
- No backups on replicas*
- Specific limitations
  - E.g. FILESTREAM, cross-database consistency of transactions

Microsoft SQL Server

AlwaysOn AG

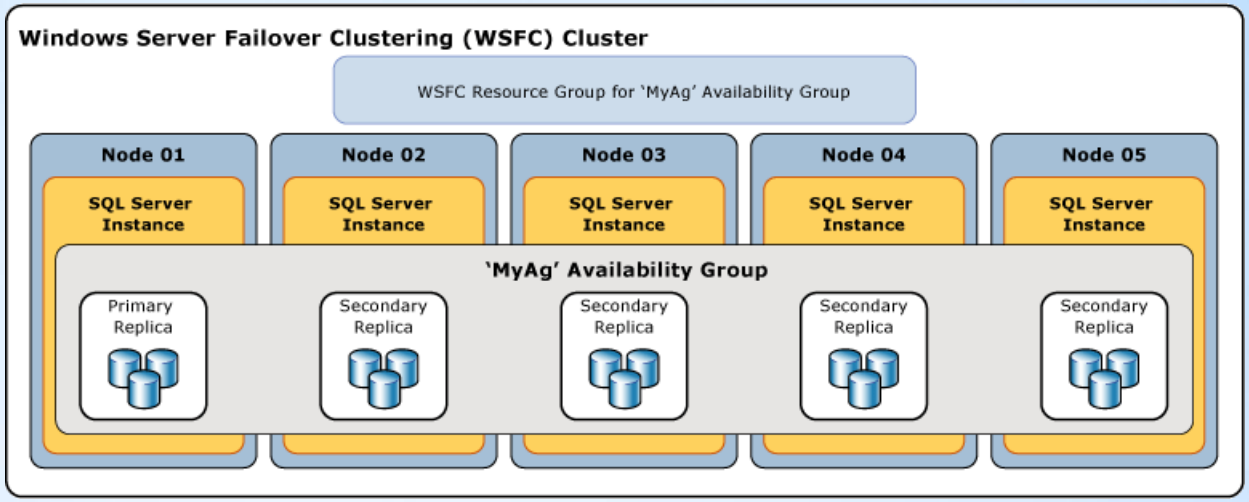
▶ AG as evolution of Database Mirroring

- Coordinated failover of multiple DBs
- Multiple replicas (up to 4)
- Synchronous/asynchronous replicas
- Up-to-date readable replicas
- Ability to off-load maintenance on replicas e.g. backups

Microsoft SQL Server

AlwaysOn AG

▶ Overview

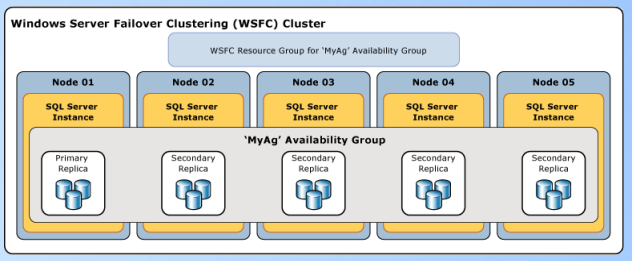


http://msdn.microsoft.com/en-us/library/ff877884.aspx

Microsoft SQL Server

AlwaysOn AG

▶ Windows Server Failover Clustering (WSFC)



- Used by Availability Groups for
  - Primary health detection
  - Inter-node health detection
  - Failover coordination
  - Distributed data store for settings and state
  - Distributed change notifications between nodes

Microsoft SQL Server

AlwaysOn AG

▶ Synchronization and Failover

- Availability Mode
  - Synchronous
  - Asynchronous
- Failover mode
  - Automatic
  - Manual
  - Forced

Microsoft SQL Server

AlwaysOn AG

▶ Conditions for Automatic Failover

- Primary and Secondary need
  - Synchronous Availability Mode configuration
  - Automatic Failover configuration
  - To be synchronized
- WFSC needs to reach a quorum
- Failover policies met

Microsoft SQL Server

AlwaysOn AG

▶ Flexible failover policies for Availability Groups

- Leverages FCI mechanism
  - sp_server_diagnostic
- Configuration in T-SQL
  - ALTER AVAILABILITY GROUP …
    SET (HEALTH_CHECK_TIMEOUT = 30000);
  - ALTER AVAILABILITY GROUP …
    SET (FAILURE_CONDITION_LEVEL = 3);

Microsoft SQL Server

AlwaysOn AG

▶ Replicas and backups

- Backup support on secondary replicas
  - Full backups with COPY_ONLY
  - Differential not supported!
  - Log backups
    - Form a single log-chain across all replicas
- Automated Backup Preference & Priority
  - No immediate effect
  - Backup scripts need to query UDF
    - sys.fn_hadr_backup_is_preferred_replica('dbname')

Microsoft SQL Server

AlwaysOn AG

▶ Read-Only Access

- Replicas roles
  - Primary: READ_WRITE | ALL
  - Secondary: NO | READ_ONLY | ALL
- New connection property
  - ApplicationIntent
  - Name slightly different across client APIs
- Read-Only Routing

Microsoft SQL Server

AlwaysOn AG

▶ Topology example 1

Direct attached storage local, regional and geo secondaries

Synchronous data movement

Asynchronous data movement

Microsoft SQL Server

AlwaysOn AG

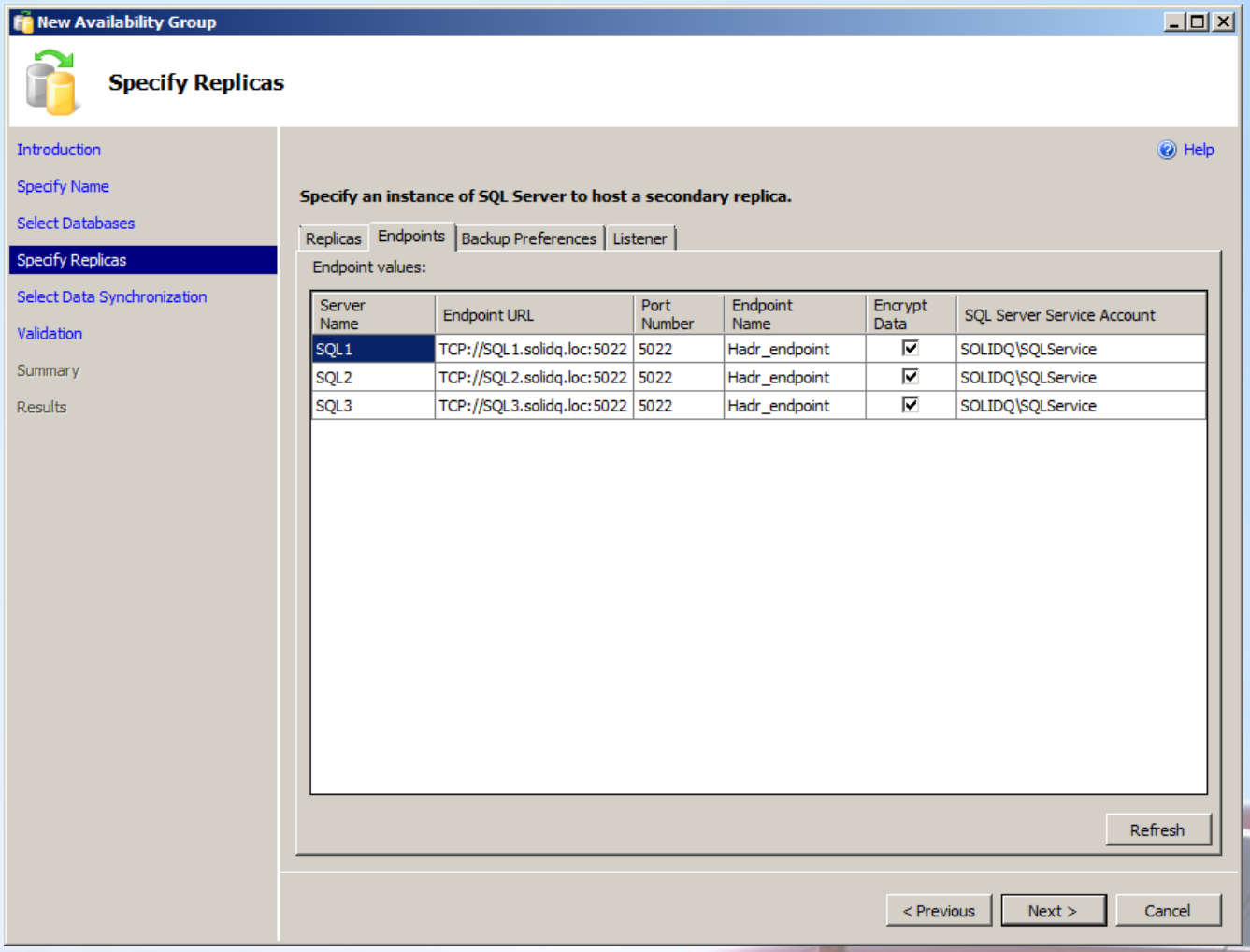
▶ Demo

- Primary Datacenter
  - Primary replica
  - Secondary replica for HA and reporting
- Secondary Datacenter
  - Secondary replica for DR

Microsoft SQL Server

Step-by-step AG

▶ AG Wizard: configure Endpoints



Microsoft SQL Server

Step-by-step AG

▶ AG Wizard: configure Backup preferences

New Availability Group

Specify Replicas

Introduction
Specify Name
Select Databases
Specify Replicas
Select Data Synchronization
Validation
Summary
Results

Help

Specify an instance of SQL Server to host a secondary replica.

Replicas | Endpoints | Backup Preferences | Listener

Where should backups occur?

- **Prefer Secondary**
  Automated backups for this availability group should occur on a secondary replica. If there is no secondary replica available, backups will be performed on the primary replica.
- **Secondary only**
  All automated backups for this availability group must occur on a secondary replica.
- **Primary**
  All automated backups for this availability group must occur on the current primary replica.
- **Any Replica**
  Backups can occur on any replica in the availability group.

Replica backup priorities:

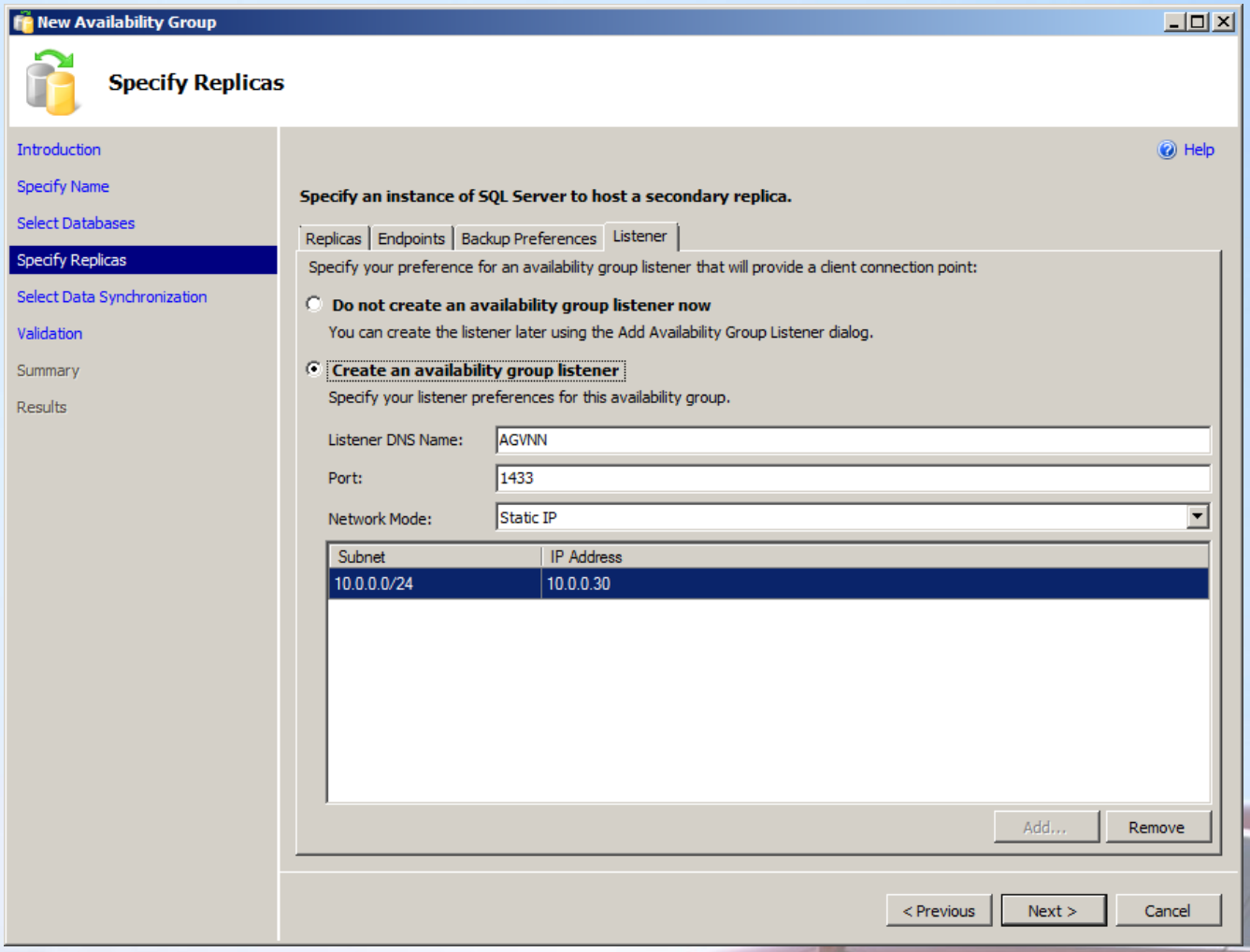
Server Instance	Backup Priority (Lowest=1, Highest=100)	Exclude Replica
SQL1	50	☐
SQL2	100	☐
SQL3	50	☑

< Previous | Next > | Cancel

Microsoft SQL Server

Step-by-step AG

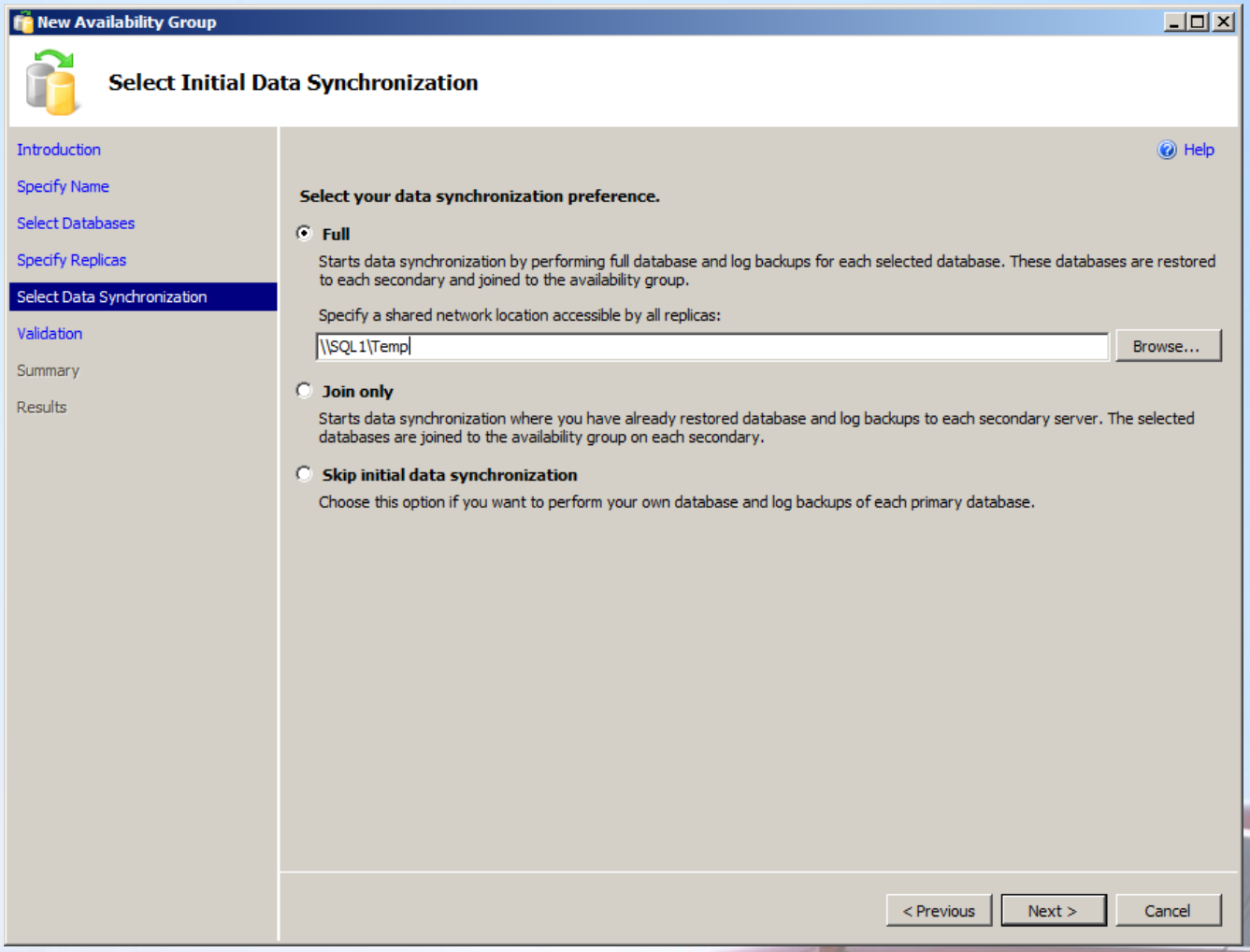
▶ AG Wizard: configure Listener



Microsoft SQL Server

Step-by-step AG

- AG Wizard: configure data synchronization



Microsoft SQL Server

Step-by-step AG

▶ AG Wizard: Validation

New Availability Group

Validation

- Introduction
- Specify Name
- Select Databases
- Specify Replicas
- Select Data Synchronization
- Validation
- Summary
- Results

Help

Results of availability group validation.

	Name	Result
✓	Checking whether the endpoint is encrypted using a compatible algorithm	Success
✓	Checking shared network location	Success
✓	Checking for free disk space on the server instance that hosts secondary replica SQL2	Success
✓	Checking if the selected databases already exist on the server instance that hosts secondary replica SQL2	Success
✓	Checking for compatibility of the database file locations on the server instance that hosts secondary replica SQL2	Success
✓	Checking for the existence of the database files on the server instance that hosts secondary replica SQL2	Success
✓	Checking for free disk space on the server instance that hosts secondary replica SQL3	Success
✓	Checking if the selected databases already exist on the server instance that hosts secondary replica SQL3	Success
✓	Checking for compatibility of the database file locations on the server instance that hosts secondary replica SQL3	Success
✓	Checking for the existence of the database files on the server instance that hosts secondary replica SQL3	Success
✓	Checking the listener configuration	Success

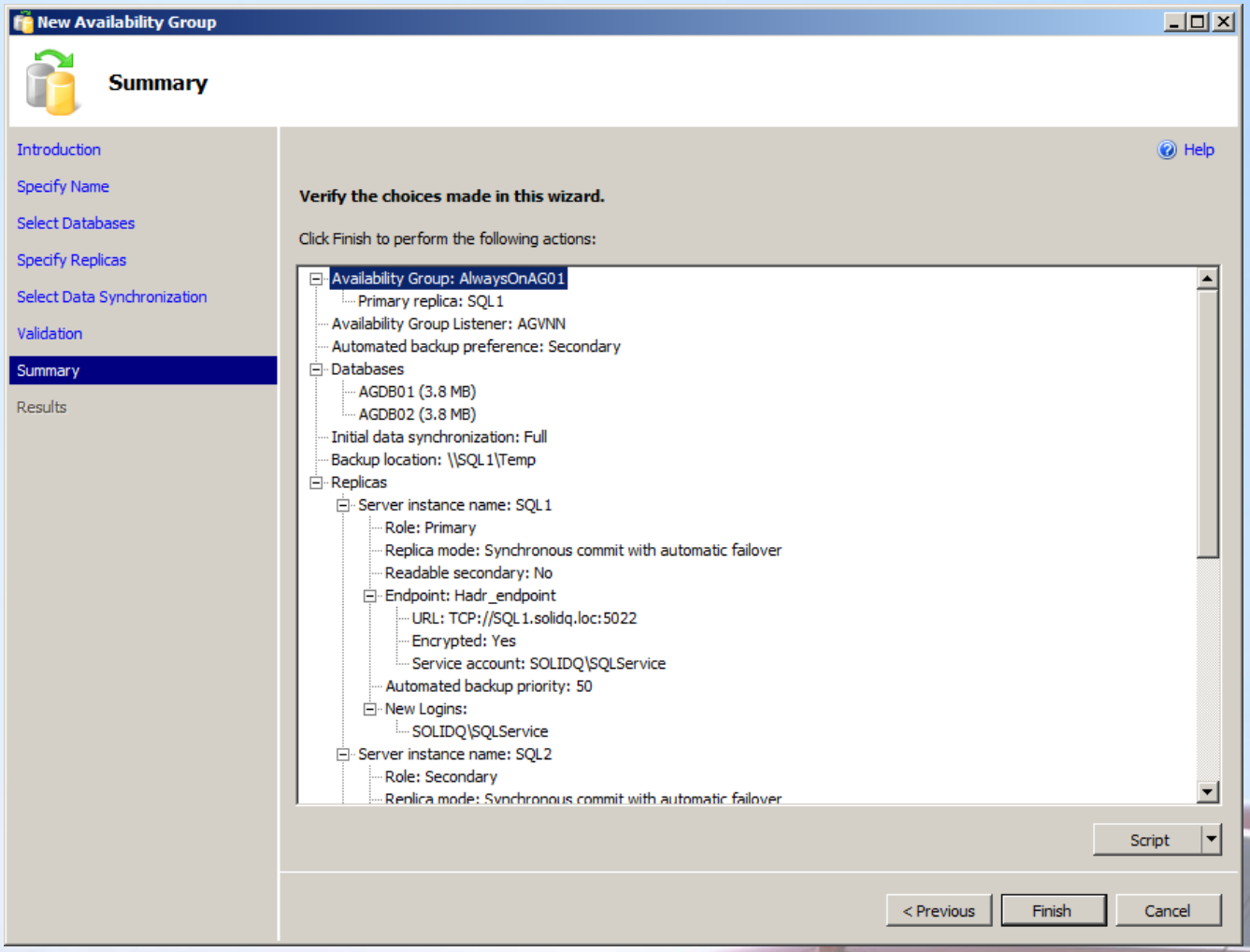
Re-run Validation

< Previous | Next > | Cancel

Microsoft SQL Server

Step-by-step AG

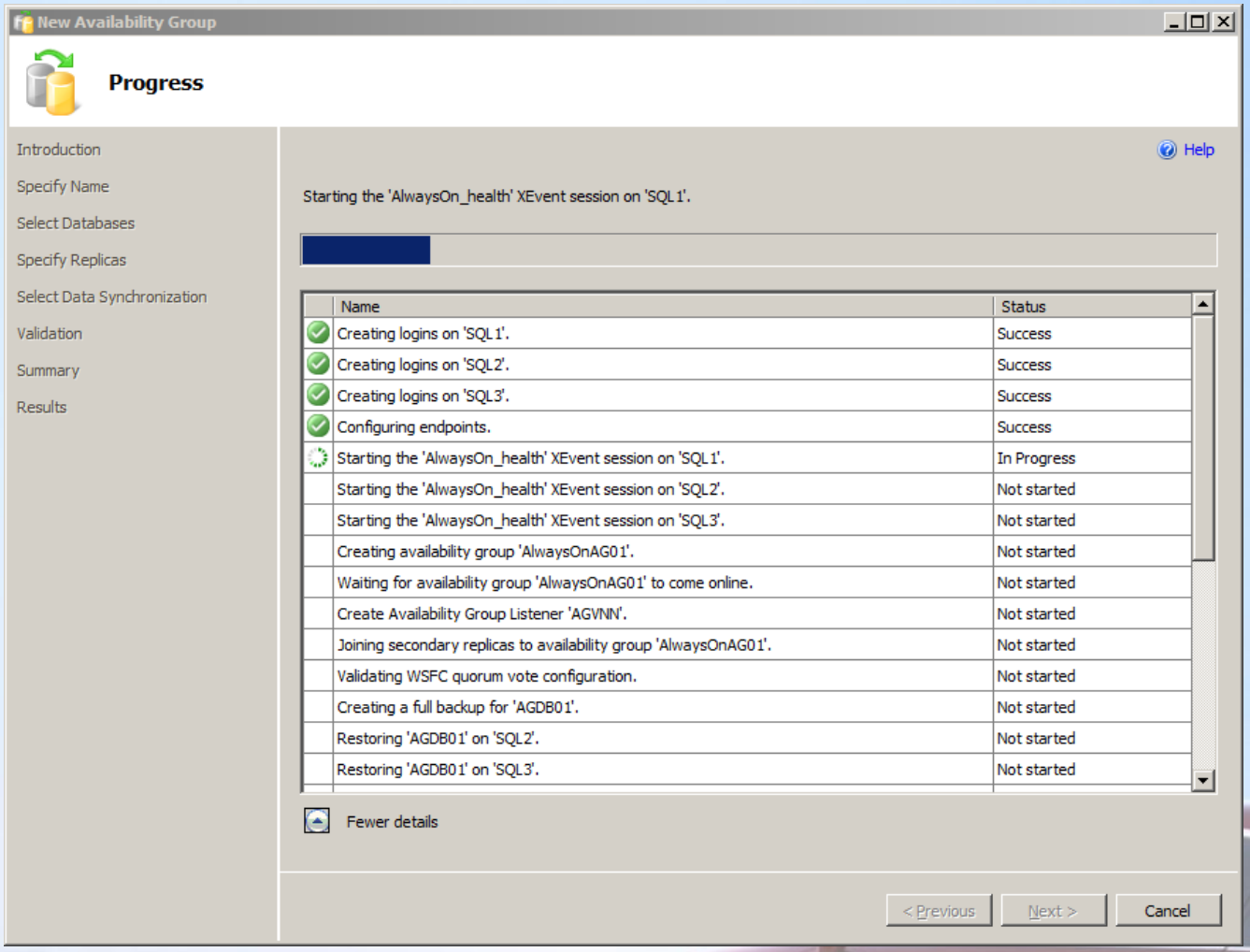
▶ AG Wizard: Summary



Microsoft SQL Server

Step-by-step AG

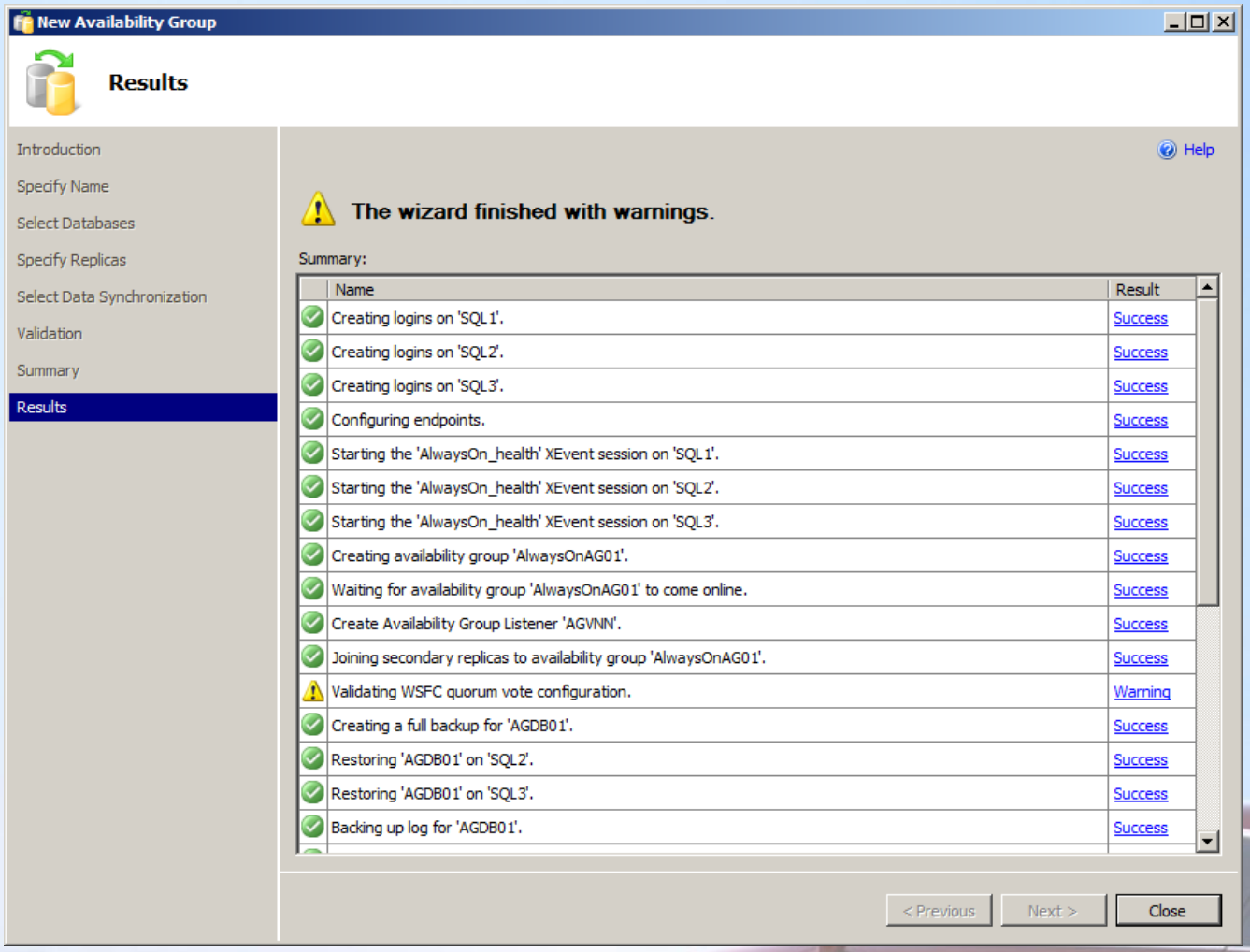
▶ AG Wizard: Progress



Microsoft SQL Server

Step-by-step AG

▶ AG Wizard: Results



Microsoft SQL Server

Step-by-step AG

- AG Dashboard

AlwaysOnAG01:SQL1 - Microsoft SQL Server Management Studio (Administrator)

File Edit View Project Debug Tools Window Help

New Query

Object Explorer

Connect

SQL1 (SQL Server 11.0.2100 - SOLIDQ\Adr
Databases
System Databases
Database Snapshots
AGDB01
AGDB02
Security
Server Objects
Replication
AlwaysOn High Availability
Availability Groups
AlwaysOnAG01 (Primary)
Availability Replicas
SQL1 (Primary)
SQL2 (Secondar
SQL3 (Secondar
Availability Databas
AGDB01
AGDB02
Availability Group Li
AGVNN
Management
Integration Services Catalogs
SQL Server Agent

Add Database...
Add Replica...
Add Listener...
Show Dashboard
Failover...
Start PowerShell
Reports
Delete...
Refresh
Properties

AlwaysOnAG01:SQL1 | Dashboard: SQL1

AlwaysOnAG01: hosted by SQL1 (Replica role: Primary)

Last updated: 3/31/2012 2:26:57 AM

Auto refresh: on

Start Failover Wizard
View AlwaysOn Health Events
View Cluster Quorum Information

Availability group state: Healthy
Primary instance: SQL1
Failover mode: Automatic
Cluster state: SQLCLUSTER (Normal Quorum)

Availability replica:

Name	Role	Failover Mode	Synchronization State	Issues
SQL1	Primary	Automatic	Synchronized	
SQL2	Secondary	Automatic	Synchronized	
[illegible]	[illegible]ndary	Manual	Synchronizing	

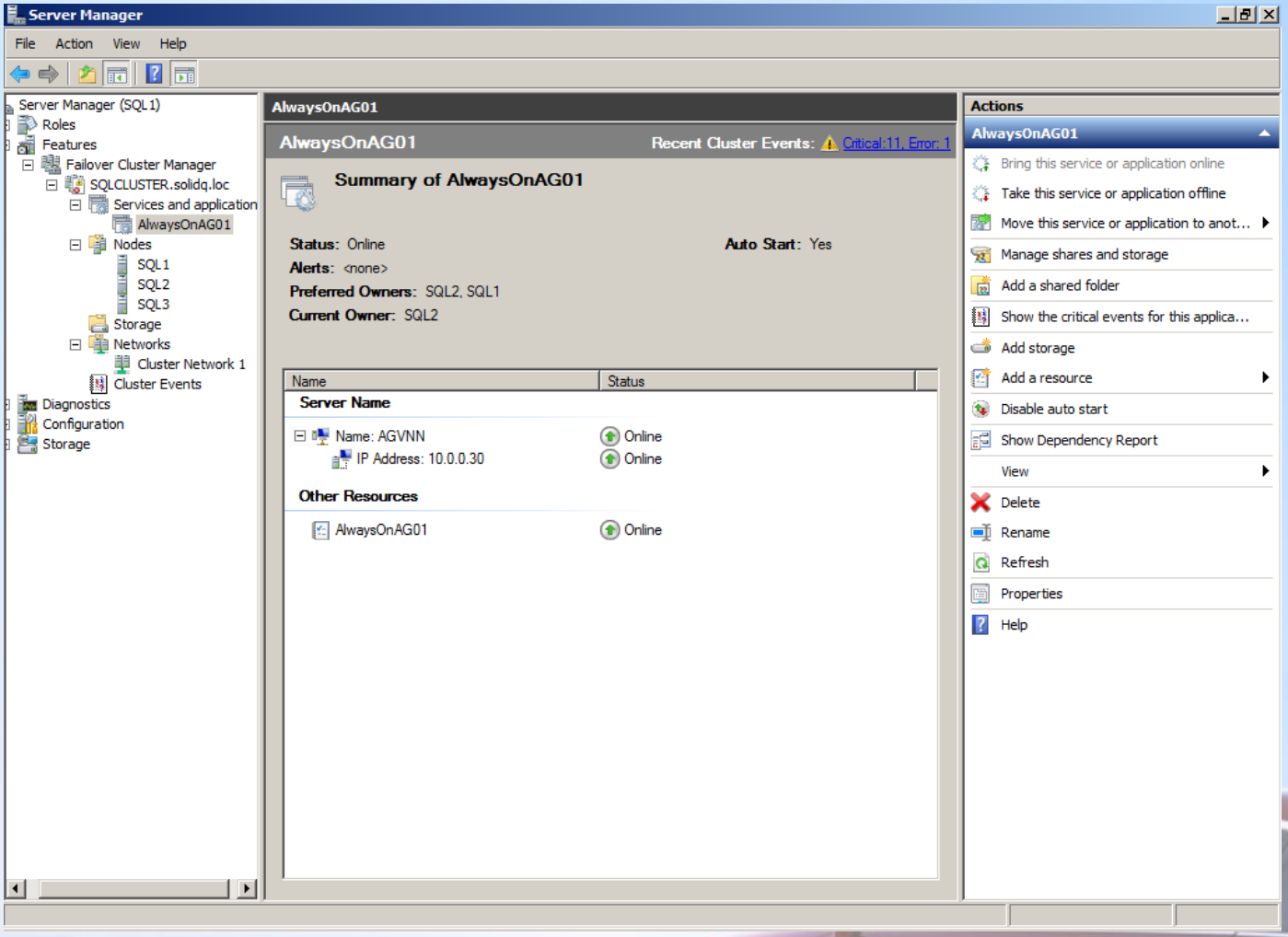
	Replica	Synchronization State	Failover Readin...	Issues
	SQL1	Synchronized	No Data Loss	
	SQL1	Synchronized	No Data Loss	
	SQL2	Synchronized	No Data Loss	
	SQL2	Synchronized	No Data Loss	
AGDB01	SQL3	Synchronizing	Data Loss	
AGDB02	SQL3	Synchronizing	Data Loss	

Ready

Microsoft SQL Server

Step-by-step AG

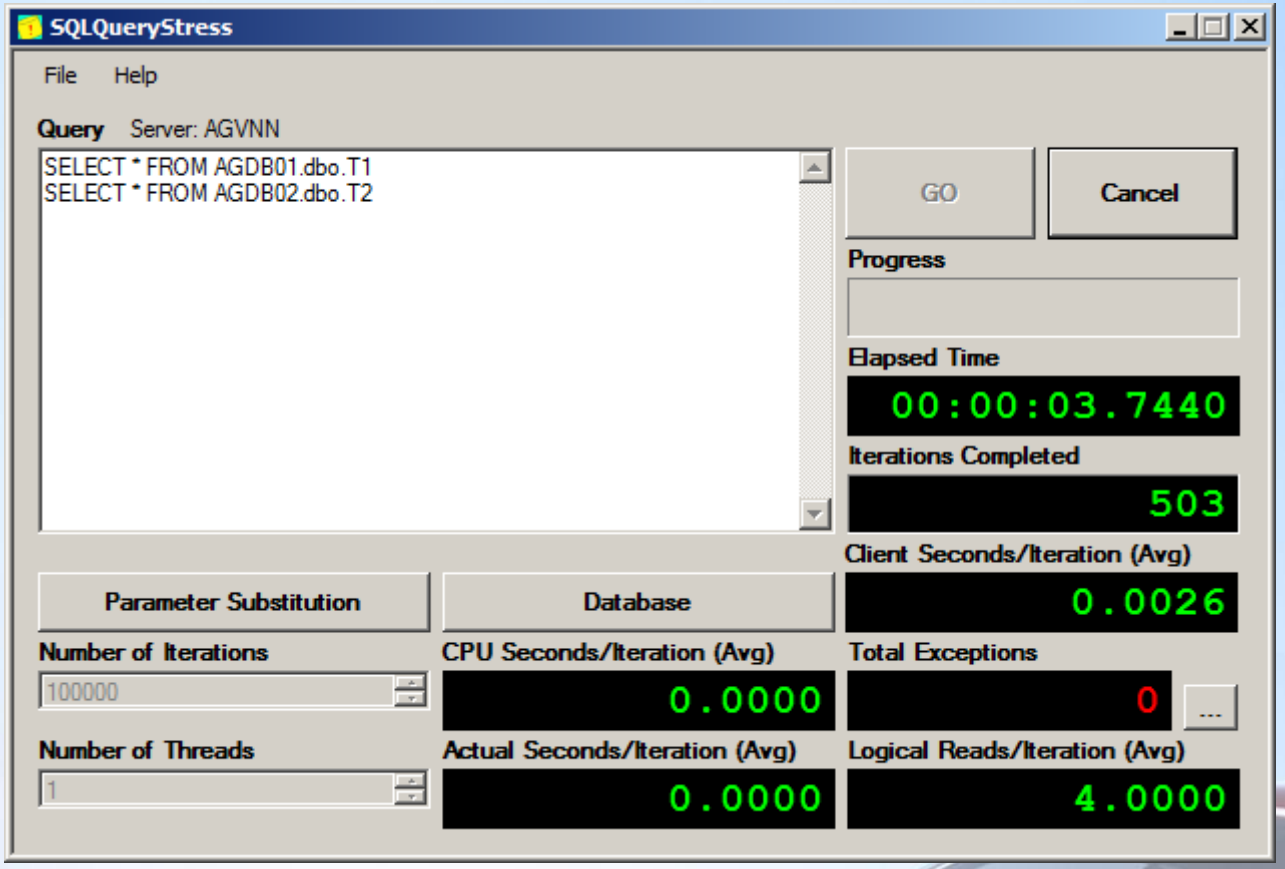
- Cluster configuration of AG after installation



Microsoft SQL Server

Step-by-step AG

- Simulate a failure: start a workload



Microsoft SQL Server

Step-by-step AG

- Simulate a failure: start a manual Failover

AlwaysOnAG01:SQL1 - Microsoft SQL Server Management Studio (Administrator)

File Edit View Project Debug Tools Window Help

New Query

Object Explorer

Connect

SQL1 (SQL Server 11.0.2100 - SOLIDQ\Adr
Databases
System Databases
Database Snapshots
AGDB01
AGDB02
Security
Server Objects
Replication
AlwaysOn High Availability
Availability Groups
AlwaysOnAG01 (Primary)
Availability Replicas
SQL1 (Primary)
SQL2 (Seconda
SQL3 (Seconda
Availability Databa
AGDB01
AGDB02
Availability Group L
AGVNN
Management
Integration Services Catalogs
SQL Server Agent

Add Database...
Add Replica...
Add Listener...
Show Dashboard
Failover...
Start PowerShell
Reports
Delete...
Refresh
Properties

AlwaysOnAG01:SQL1 | Dashboard: SQL1

AlwaysOnAG01: hosted by SQL1 (Replica role: Primary)

Last updated: 3/31/2012 2:16:54 PM
Auto refresh: on

Availability group state: Healthy
Primary instance: SQL1
Failover mode: Automatic
Cluster state: SQLCLUSTER (Normal Quorum)

Start Failover Wizard
View AlwaysOn Health Events
View Cluster Quorum Information

Availability replica:

Name	Role	Failover Mode	Synchronization State	Issues
SQL1	Primary	Automatic	Synchronized	
SQL2	Second...	Automatic	Synchronized	
	Second...	Manual	Synchronizing	

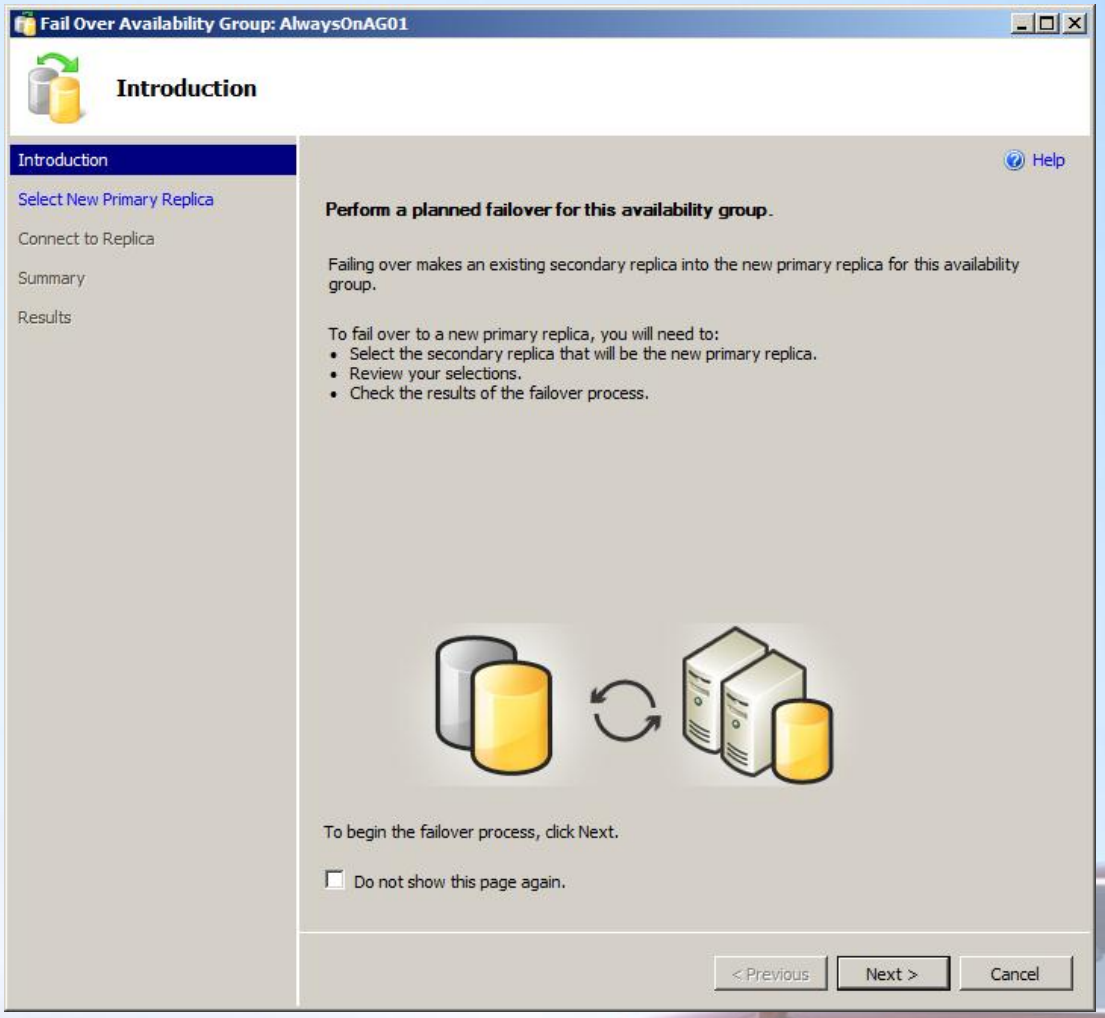
Replica	Synchronization State	Failover Readin...	Issues
SQL1	Synchronized	No Data Loss	
SQL1	Synchronized	No Data Loss	
SQL2	Synchronized	No Data Loss	
SQL2	Synchronized	No Data Loss	
AGDB01 SQL3	Synchronizing	Data Loss	
AGDB02 SQL3	Synchronizing	Data Loss	

Ready

Microsoft SQL Server

Step-by-step AG

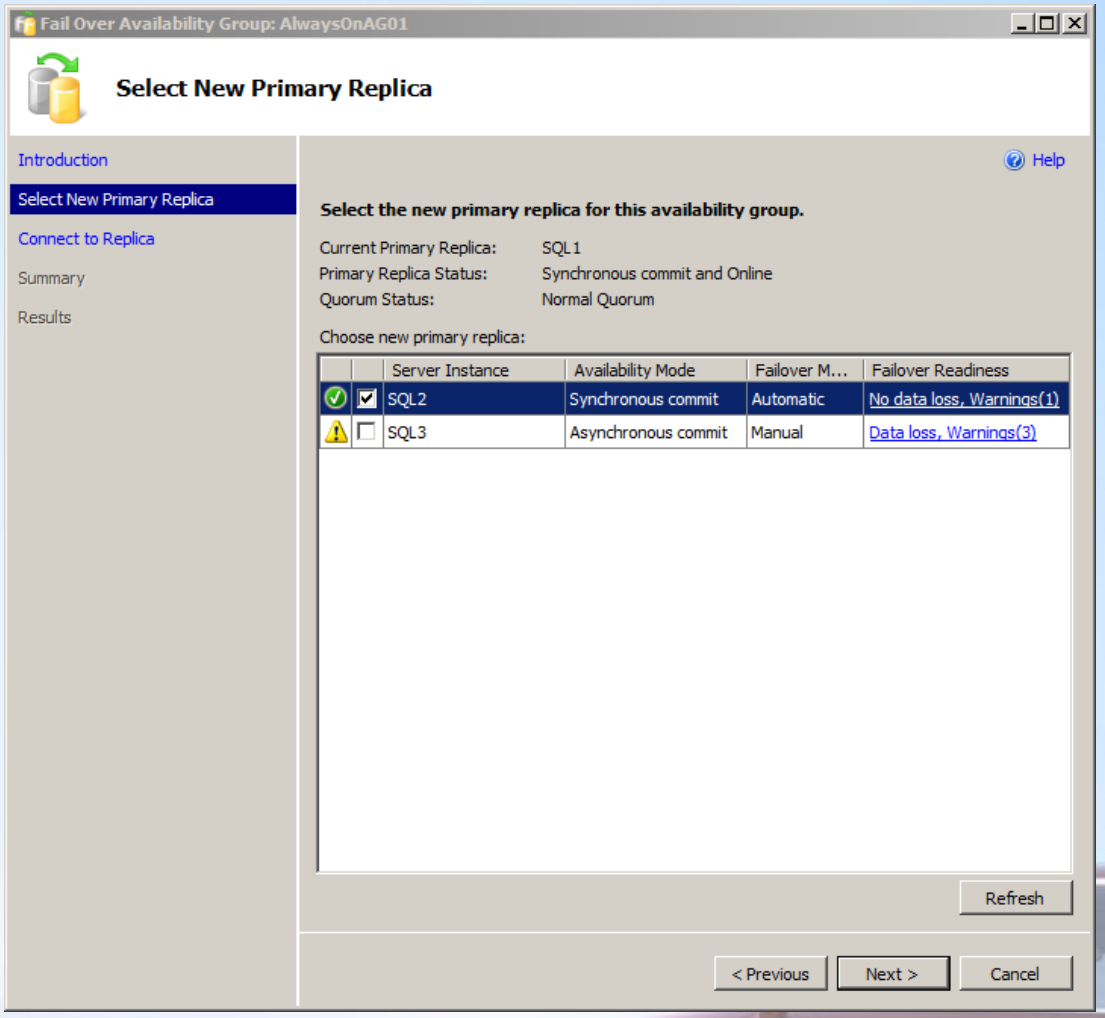
▶ Failover Wizard: Introduction



Microsoft SQL Server

Step-by-step AG

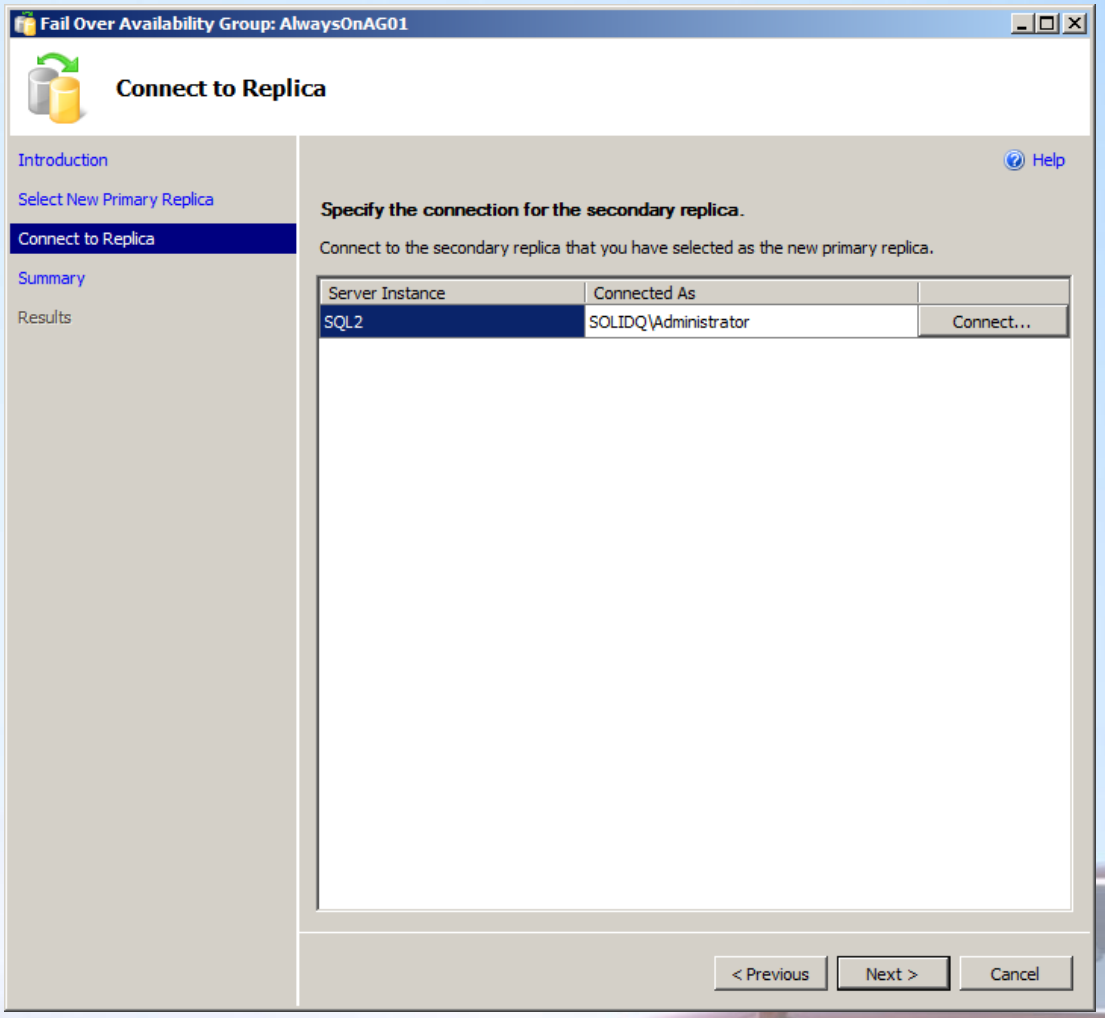
- Failover Wizard: select new Primary Replica



Microsoft SQL Server

Step-by-step AG

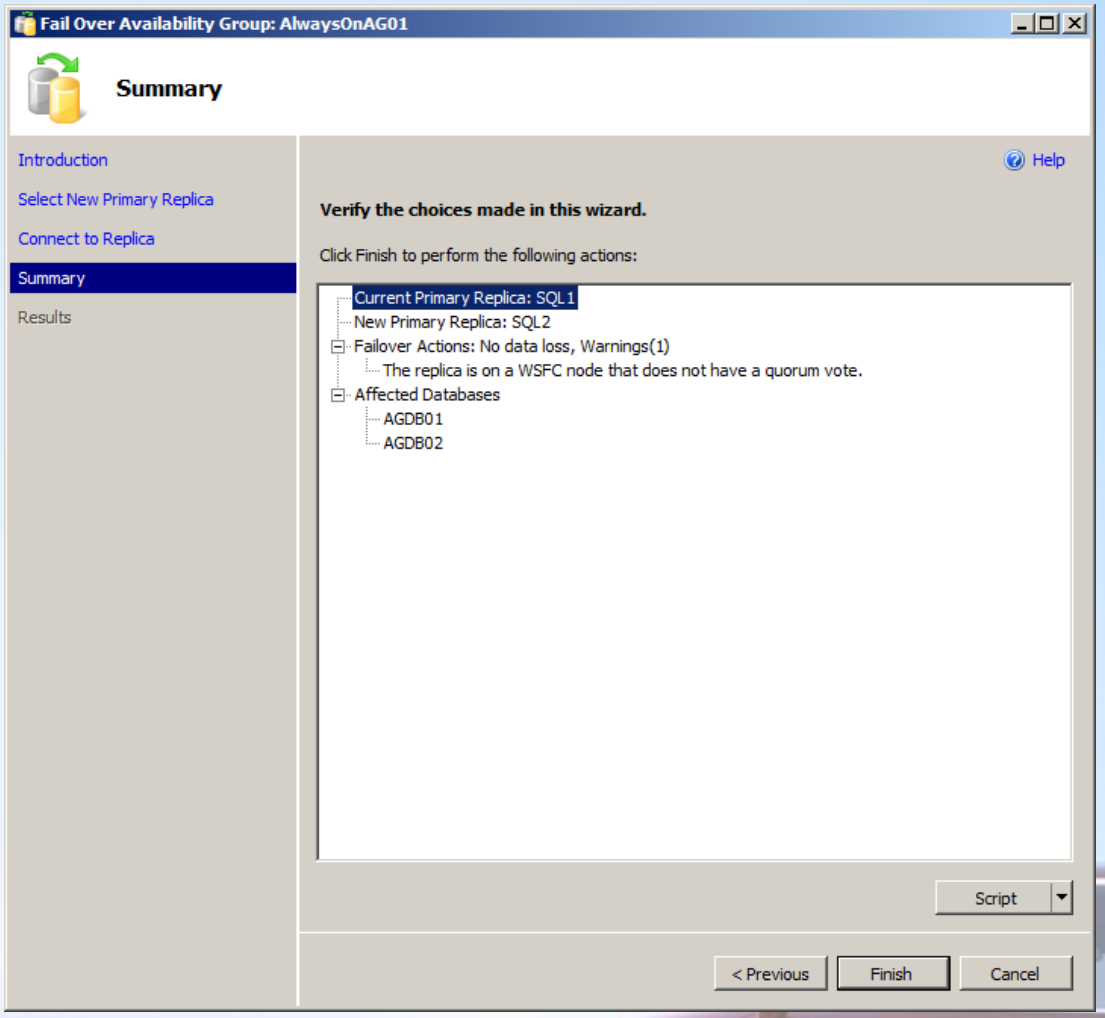
- Failover Wizard: connect to Secondary Replica



Microsoft SQL Server

Step-by-step AG

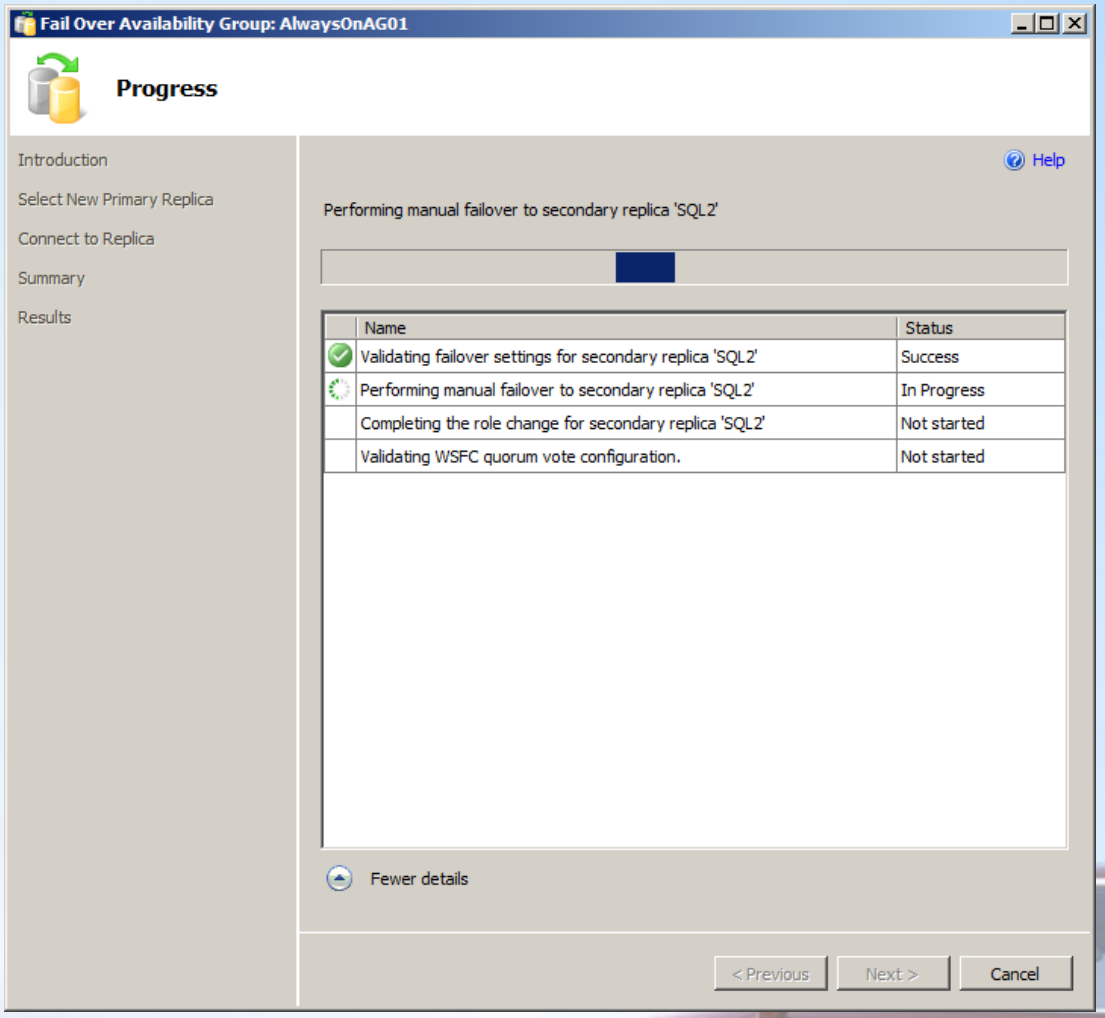
- Failover Wizard: Summary



Microsoft SQL Server

Step-by-step AG

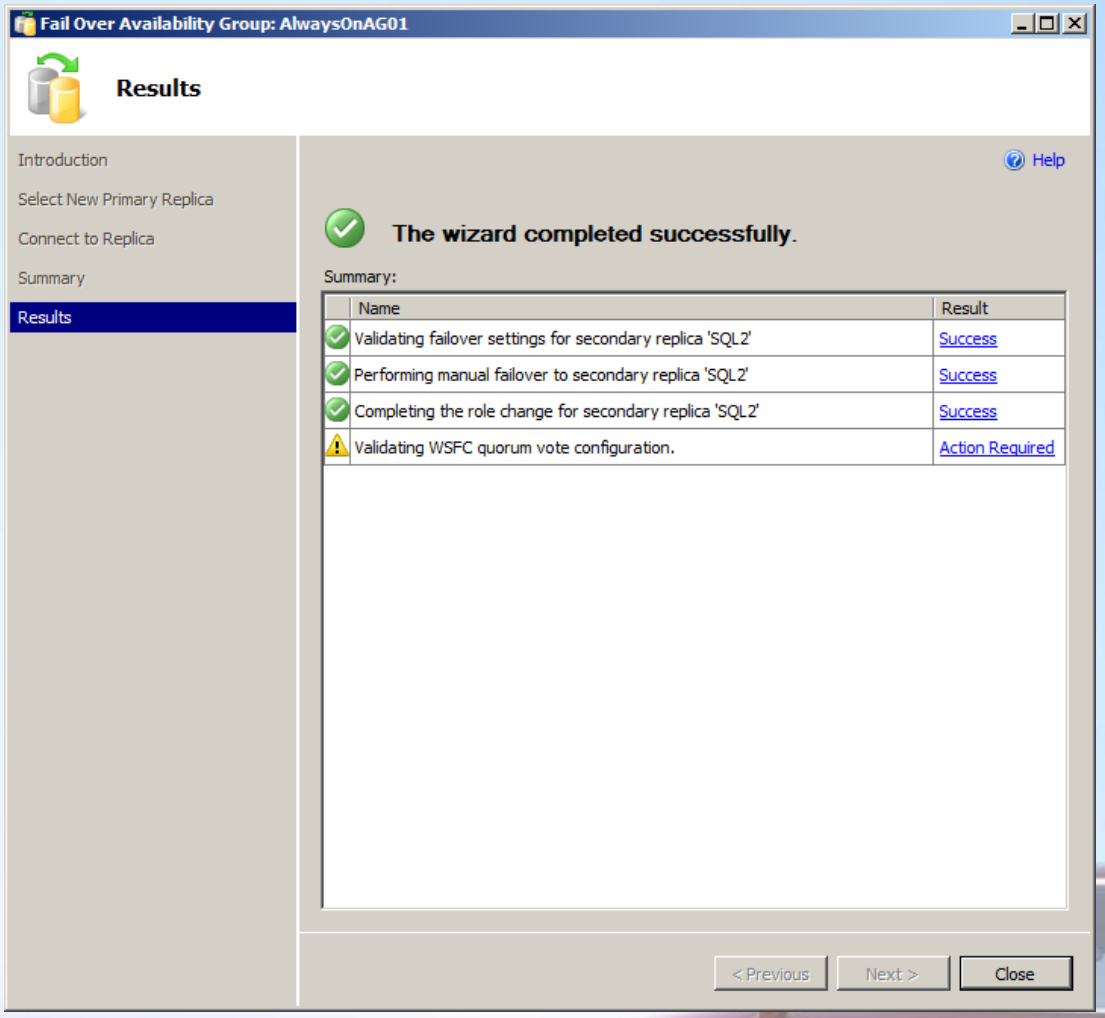
- Failover Wizard: Progress



Microsoft SQL Server

Step-by-step AG

- Failover Wizard: Complete



Microsoft SQL Server

Step-by-step AG

- Simulate a failure: Dashboard on new Secondary Replica

AlwaysOnAG01:SQL1 - Microsoft SQL Server Management Studio (Administrator)

File Edit View Project Debug Tools Window Help

New Query

Object Explorer

Connect

- SQL1 (SQL Server 11.0.2100 - SOLIDQ\Adr
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01
    - AGDB02
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
    - Availability Groups
      - AlwaysOnAG01 (Secondary)
        - Availability Replicas
          - SQL1 (Secondary)
          - SQL2
          - SQL3
        - Availability Databases
          - AGDB01
          - AGDB02
        - Availability Group Listeners
          - AGVNN
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

AlwaysOnAG01:SQL1 | Dashboard: SQL1

AlwaysOnAG01: hosted by SQL1 (Replica role: Secondary)

Last updated: 3/31/2012 2:20:54 PM

Auto refresh: on

Availability group state: No data available on secondary replicas.

Primary instance: SQL2

Failover mode: Automatic

Cluster state: SQLCLUSTER (Normal Quorum)

Start Failover Wizard

View AlwaysOn Health Events

View Cluster Quorum Information

Availability replica:

Name	Role	Failover Mode	Synchronization State	Issues
SQL1	Second...	Automatic	Synchronized	

Group by

Name	Replica	Synchronization State	Failover Readin...	Issues
SQL1				
AGDB01	SQL1	Synchronized	No Data Loss	
AGDB02	SQL1	Synchronized	No Data Loss	

Ready

Microsoft SQL Server

Step-by-step AG

- Simulate a failure: Dashboard on new Primary Replica

AlwaysOnAG01:SQL2 - Microsoft SQL Server Management Studio (Administrator)

File Edit View Project Debug Tools Window Help

New Query

Object Explorer

Connect

- SQL1 (SQL Server 11.0.2100 - SOLIDQ\Adr
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01
    - AGDB02
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
    - Availability Groups
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        - Availability Replicas
          - SQL1 (Secondary)
          - SQL2
          - SQL3
        - Availability Databases
          - AGDB01
          - AGDB02
        - Availability Group Listeners
          - AGVNN
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

AlwaysOnAG01:SQL2 | AlwaysOnAG01:SQL1 | Dashboard: SQL1

AlwaysOnAG01: hosted by SQL2 (Replica role: Primary)

Last updated: 3/31/2012 2:21:41 PM

Auto refresh: on

Availability group state: Healthy

Primary instance: SQL2

Failover mode: Automatic

Cluster state: SQLCLUSTER (Normal Quorum)

Start Failover Wizard

View AlwaysOn Health Events

View Cluster Quorum Information

Availability replica:

Name	Role	Failover Mode	Synchronization State	Issues
SQL1	Second...	Automatic	Synchronized	
SQL2	Primary	Automatic	Synchronized	
SQL3	Second...	Manual	Synchronizing	

Group by

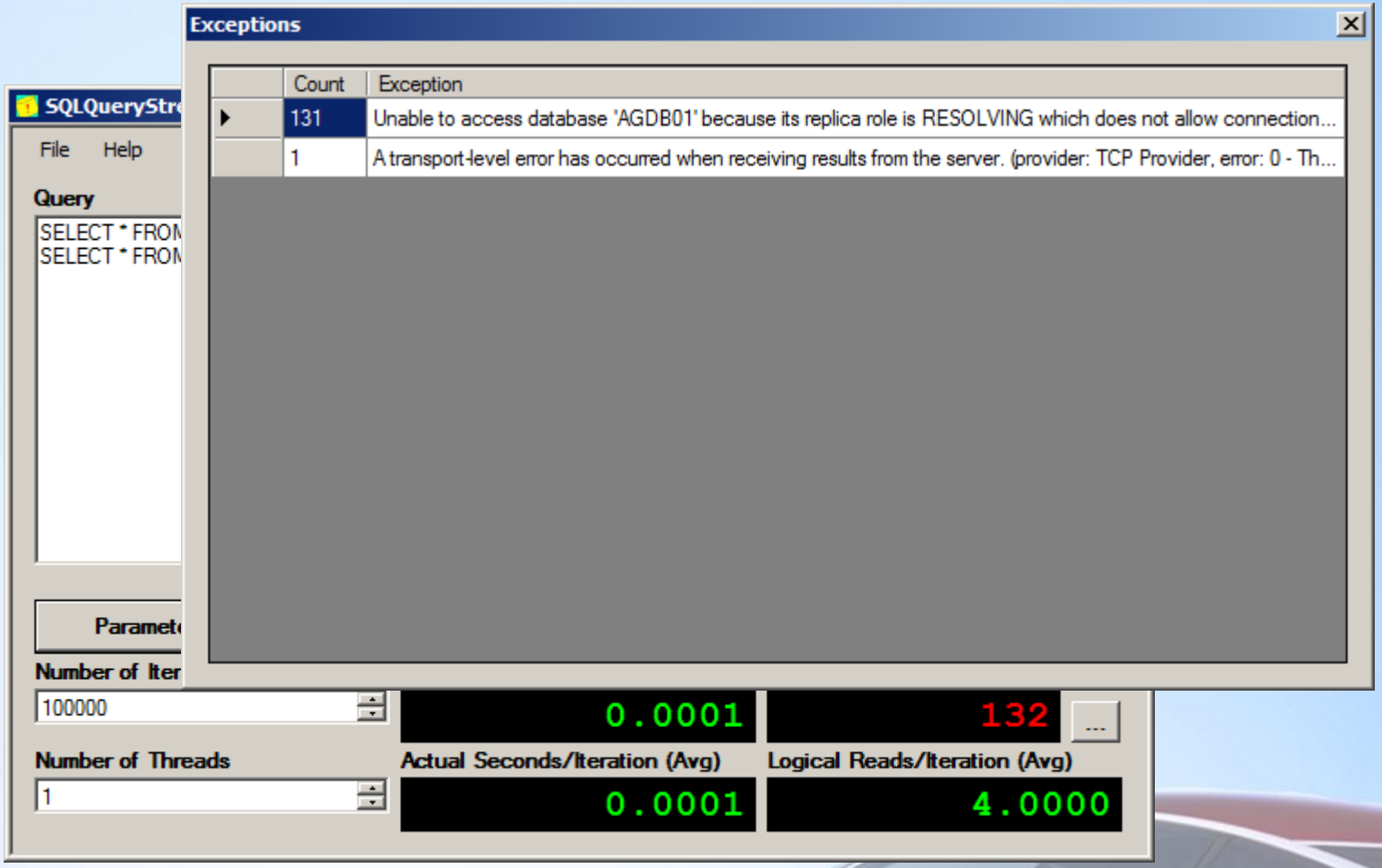
Name	Replica	Synchronization State	Failover Readin...	Issues
SQL1				
AGDB01	SQL1	Synchronized	No Data Loss	
AGDB02	SQL1	Synchronized	No Data Loss	
SQL2				
AGDB01	SQL2	Synchronized	No Data Loss	
AGDB02	SQL2	Synchronized	No Data Loss	
SQL3				
AGDB01	SQL3	Synchronizing	Data Loss	
AGDB02	SQL3	Synchronizing	Data Loss	

Ready

Microsoft SQL Server

Step-by-step AG

- Simulate a failure: clients exceptions



Microsoft SQL Server

Step-by-step AG

- Accessing AG: using the Listener's VNN

simple-queries.sql - AGVNN.master (SOLIDQ\Administrator (59)) - Microsoft SQL Server Management Studio (Administrator)

File Edit View Query Project Debug Tools Window Help

New Query | master | Execute | Debug

Object Explorer

Connect

- AGVNN (SQL Server 11.0.2100 - S
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01 (Synchronized)
    - AGDB02 (Synchronized)
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

simple-queries.sql -...DQ\Administrator (59))

```
SELECT  @@SERVERNAME AS Instance;
GO
SELECT * FROM AGDB01.dbo.T1;
SELECT * FROM AGDB02.dbo.T2;
GO
```

100 %

Results

```
Instance
--------------------------------------------------------------------------------
SQL1

(1 row(s) affected)

F1
-----------
1
2
3
4
5

(5 row(s) affected)

F1
-----------
1
2
3
4
5

(5 row(s) affected)
```

100 %

Query executed successfully. | AGVNN (11.0 RTM) | SOLIDQ\Administrator (59) | master | 00:00:00 | 11 rows

Item(s) Saved | Ln 5 | Col 3 | Ch 3 | INS

Microsoft SQL Server

Step-by-step AG

- Accessing AG: connecting directly to Secondary Replica

Step-by-step AG

- Accessing AG: Secondary Replica not accessible

simple-queries.sql - SQL2.master (SOLIDQ\Administrator (52)) - Microsoft SQL Server Management Studio (Administrator)

File Edit View Query Project Debug Tools Window Help

New Query | master | Execute | Debug

Object Explorer

Connect

- AGVNN (SQL Server 11.0.2100 - S
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01 (Synchronized)
    - AGDB02 (Synchronized)
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

simple-queries.sql -...DQ\Administrator (52))

```
SELECT  @@SERVERNAME AS Instance;
GO
SELECT * FROM AGDB01.dbo.T1;
SELECT * FROM AGDB02.dbo.T2;
GO
```

100 %

Results

```
Instance
--------------------------------------------------------------------------------
SQL2

(1 row(s) affected)

Msg 976, Level 14, State 1, Line 1
The target database, 'AGDB01', is participating in an availability group and is currently not accessible
for queries. Either data movement is suspended or the availability replica is not enabled for read access.
To allow read-only access to this and other databases in the availability group, enable read access to
one or more secondary availability replicas in the group.

For more information, see the ALTER AVAILABILITY GROUP statement in SQL Server Books Online.
```

100 %

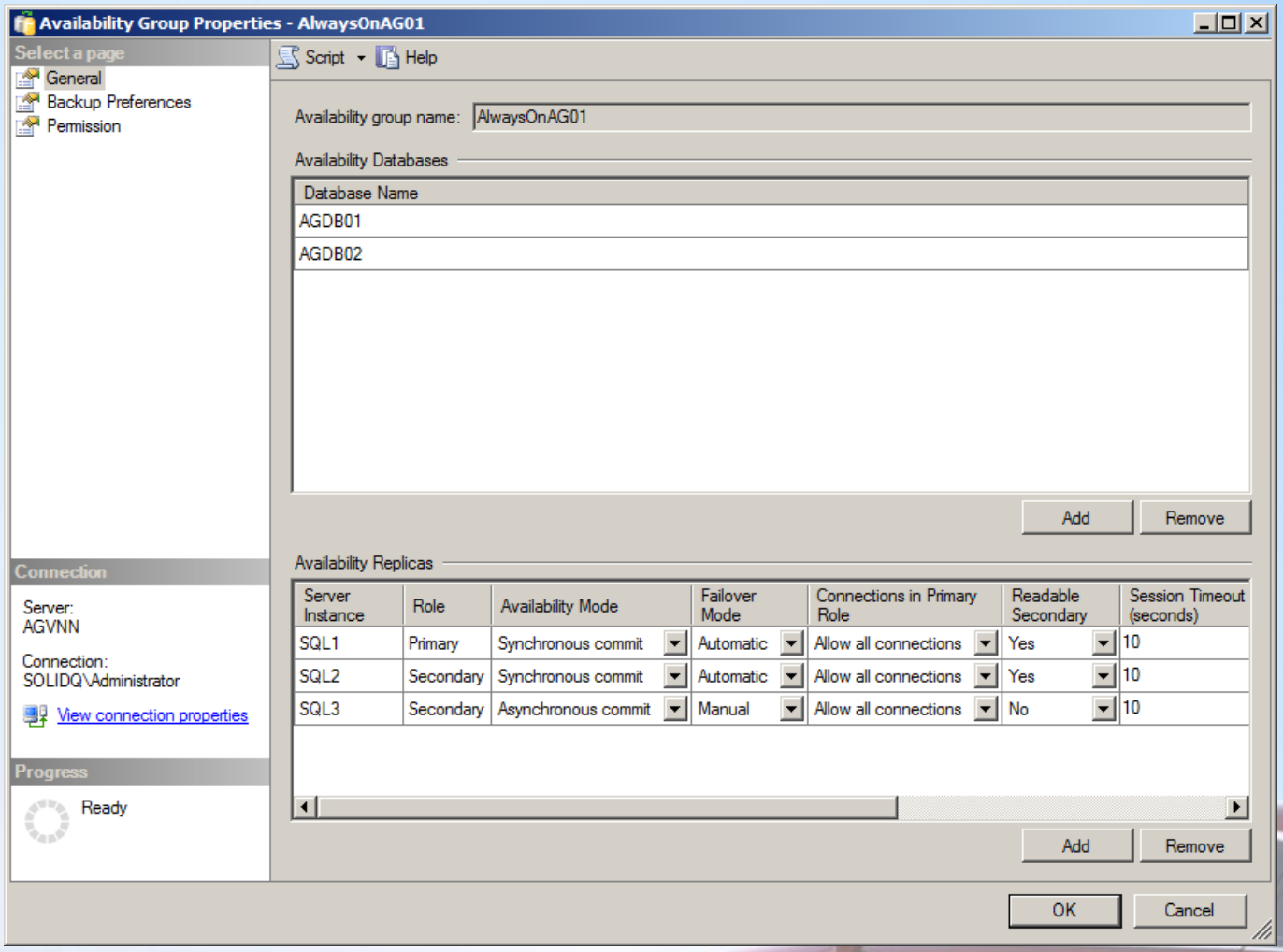
Query completed with errors. | SQL2 (11.0 RTM) | SOLIDQ\Administrator (52) | master | 00:00:00 | 1 rows

Item(s) Saved | Ln 14 | Col 1 | Ch 1 | INS

Microsoft SQL Server

Step-by-step AG

- Accessing AG: configure readable Secondary Replica



Microsoft SQL Server

Step-by-step AG

- Accessing AG: readable Secondary Replica

simple-queries.sql - SQL2.master (SOLIDQ\Administrator (52)) - Microsoft SQL Server Management Studio (Administrator)

File Edit View Query Project Debug Tools Window Help

New Query | master | Execute | Debug

Object Explorer

Connect

- AGVNN (SQL Server 11.0.2100 - S
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01 (Synchronized)
    - AGDB02 (Synchronized)
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
    - Availability Groups
      - AlwaysOnAG01 (Prima
        - Availability Replica
        - Availability Databa
        - Availability Group I
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

simple-queries.sql -...DQ\Administrator (52))

```
SELECT  @@SERVERNAME AS Instance;
GO
SELECT * FROM AGDB01.dbo.T1;
SELECT * FROM AGDB02.dbo.T2;
GO
```

100 %

Results

```
Instance
--------------------------------------------------------------------------------
SQL2

(1 row(s) affected)

F1
-----------
1
2
3
4
5

(5 row(s) affected)

F1
-----------
1
2
3
4
5

(5 row(s) affected)
```

100 %

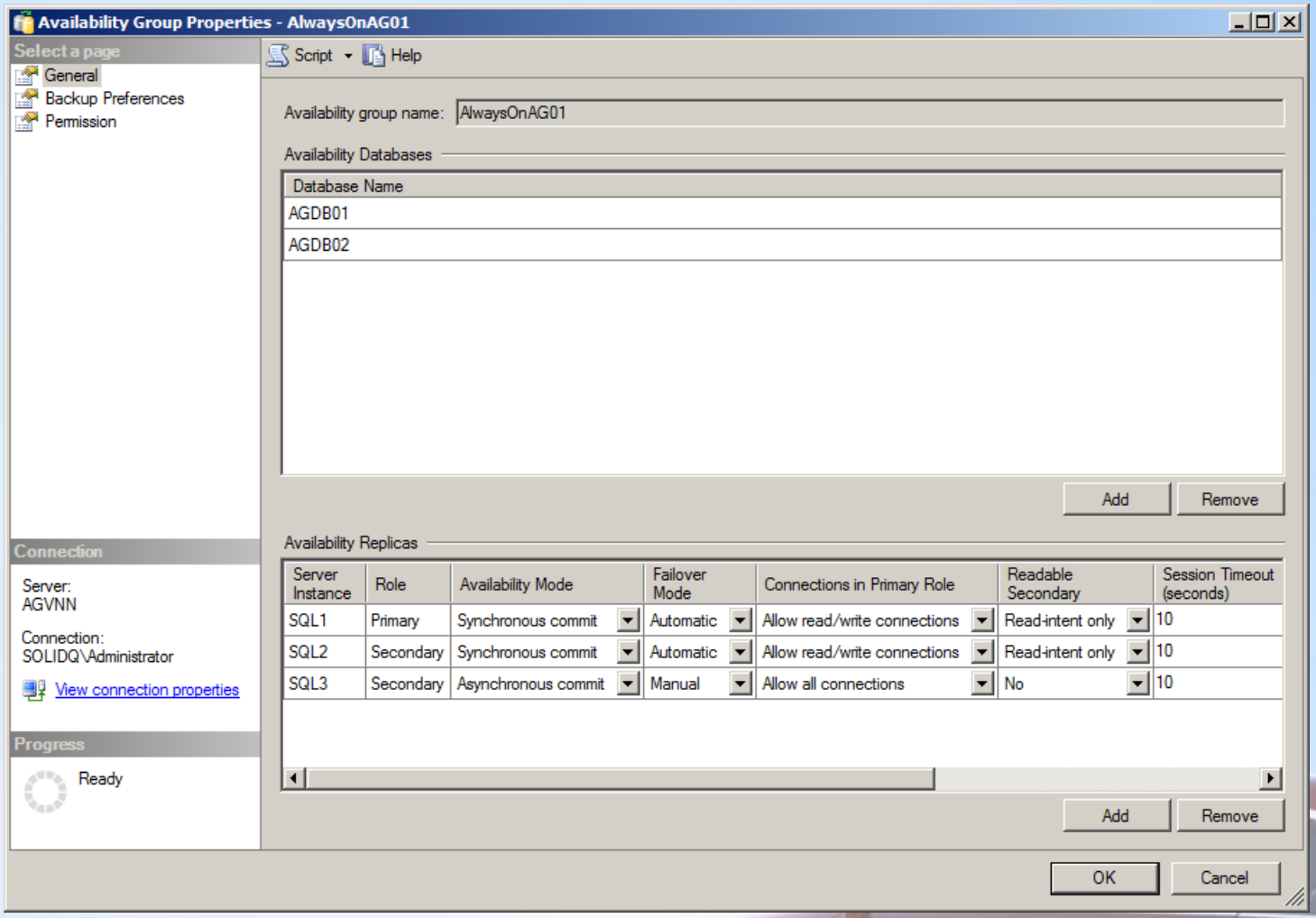
Query executed successfully. | SQL2 (11.0 RTM) | SOLIDQ\Administrator (52) | master | 00:00:00 | 11 rows

Ready | Ln 5 | Col 3 | Ch 3 | INS

Microsoft SQL Server

Step-by-step AG

- Accessing AG: configure Application Intent



Microsoft SQL Server

Step-by-step AG

- Accessing AG: VNN without Application Intent

Step-by-step AG

- Accessing AG: VNN without Application Intent defaults to read/write

simple-queries.sql - AGVNN.master (SOLIDQ\Administrator (59)) - Microsoft SQL Server Management Studio (Administrator)

File Edit View Query Project Debug Tools Window Help

New Query | master | Execute | Debug

Object Explorer

Connect

- AGVNN (SQL Server 11.0.2100 - S
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01 (Synchronized)
    - AGDB02 (Synchronized)
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
    - Availability Groups
      - AlwaysOnAG01 (Prima
        - Availability Replica
        - Availability Databa
        - Availability Group I
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

simple-queries.sql -...DQ\Administrator (59))

```
SELECT  @@SERVERNAME AS Instance;
GO
SELECT * FROM AGDB01.dbo.T1;
SELECT * FROM AGDB02.dbo.T2;
GO
```

100 %

Results

```
Instance
--------------------------------------------------------------------------------
SQL1

(1 row(s) affected)

F1
-----------
1
2
3
4
5

(5 row(s) affected)

F1
-----------
1
2
3
4
5

(5 row(s) affected)
```

100 %

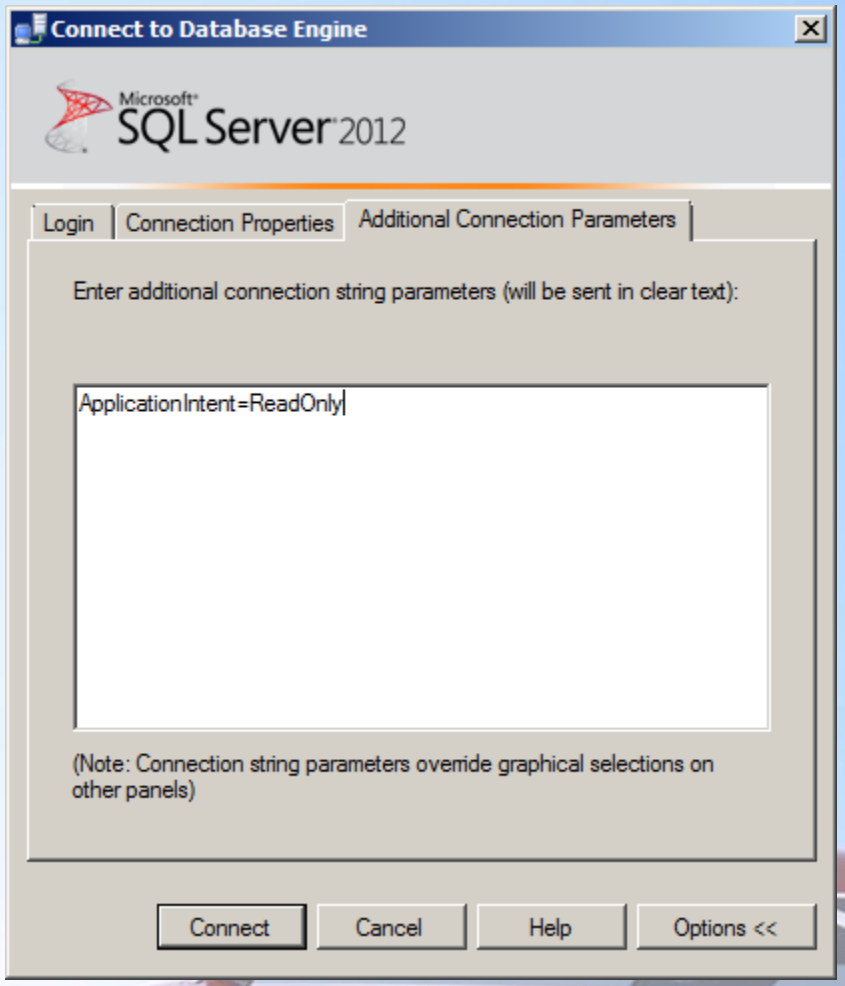
Query executed successfully. | AGVNN (11.0 RTM) | SOLIDQ\Administrator (59) | master | 00:00:00 | 11 rows

Ready | Ln 4 | Col 29 | Ch 29 | INS

Microsoft SQL Server

Step-by-step AG

- Accessing AG: VNN with Read Only Application Intent



Microsoft SQL Server

Step-by-step AG

- Accessing AG: VNN with Read Only Application Intent still on Primary

simple-queries.sql - AGVNN.master (SOLIDQ\Administrator (61)) - Microsoft SQL Server Management Studio (Administrator)

File Edit View Query Project Debug Tools Window Help

New Query | master | Execute | Debug

Object Explorer

Connect

- AGVNN (SQL Server 11.0.2100 - S
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01 (Synchronized)
    - AGDB02 (Synchronized)
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
    - Availability Groups
      - AlwaysOnAG01 (Prima
        - Availability Replica
        - Availability Databa
        - Availability Group I
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

simple-queries.sql -...DQ\Administrator (61))

```
SELECT  @@SERVERNAME AS Instance;
GO
SELECT * FROM AGDB01.dbo.T1;
SELECT * FROM AGDB02.dbo.T2;
GO
```

100 %

Results

```
Instance
--------------------------------------------------------------------------------
SQL1

(1 row(s) affected)

Msg 979, Level 14, State 1, Line 1
The target database ('AGDB01') is in an availability group and currently does not allow
read only connections.

For more information about application intent, see SQL Server Books Online.
```

100 %

Query completed with errors. | AGVNN (11.0 RTM) | SOLIDQ\Administrator (61) | master | 00:00:00 | 1 rows

Ready | Ln 11 | Col 1 | Ch 1 | INS

Microsoft SQL Server

Step-by-step AG

- Accessing AG: Read Only Routing configuration

configure-replicas.sql - SQL1.master (SOLIDQ\Administrator (62)) - Microsoft SQL Server Management Studio (Administrator)

```
-- Configure Read Only Routing URLs
ALTER AVAILABILITY GROUP [AlwaysOnAG01]
MODIFY REPLICA ON N'SQL1' WITH (SECONDARY_ROLE(READ_ONLY_ROUTING_URL=N'TCP://sql1.solidq.loc:1433'))
GO
ALTER AVAILABILITY GROUP [AlwaysOnAG01]
MODIFY REPLICA ON N'SQL2' WITH (SECONDARY_ROLE(READ_ONLY_ROUTING_URL=N'TCP://sql2.solidq.loc:1433'))
GO

-- Configure Read Only Routing lists
ALTER AVAILABILITY GROUP [AlwaysOnAG01]
MODIFY REPLICA ON N'SQL1' WITH (PRIMARY_ROLE(READ_ONLY_ROUTING_LIST=(N'SQL2', N'SQL1')))
GO
ALTER AVAILABILITY GROUP [AlwaysOnAG01]
MODIFY REPLICA ON N'SQL2' WITH (PRIMARY_ROLE(READ_ONLY_ROUTING_LIST=(N'SQL1', N'SQL2')))
GO
```

```
Command(s) completed successfully.
```

Query executed successfully. SQL1 (11.0 RTM) SOLIDQ\Administrator (62) master 00:00:01 0 rows

Microsoft SQL Server

Step-by-step AG

- Accessing AG: VNN with Read Only Application Intent again on Primary

simple-queries.sql - AGVNN.master (SOLIDQ\Administrator (61)) - Microsoft SQL Server Management Studio (Administrator)

File Edit View Query Project Debug Tools Window Help

New Query | master | Execute | Debug

Object Explorer

Connect

- AGVNN (SQL Server 11.0.2100 - S
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01 (Synchronized)
    - AGDB02 (Synchronized)
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
    - Availability Groups
      - AlwaysOnAG01 (Prima
        - Availability Replica
        - Availability Databa
        - Availability Group I
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

simple-queries.sql -...DQ\Administrator (61))

```
SELECT  @@SERVERNAME AS Instance;
GO
SELECT * FROM AGDB01.dbo.T1;
SELECT * FROM AGDB02.dbo.T2;
GO
```

100 %

Results

```
Instance
--------------------------------------------------------------------------------
SQL1

(1 row(s) affected)

Msg 979, Level 14, State 1, Line 1
The target database ('AGDB01') is in an availability group and currently does not allow
read only connections.

For more information about application intent, see SQL Server Books Online.
```

100 %

Query completed with errors. | AGVNN (11.0 RTM) | SOLIDQ\Administrator (61) | master | 00:00:00 | 1 rows

Ready | Ln 11 | Col 1 | Ch 1 | INS

Microsoft SQL Server

Step-by-step AG

- Accessing AG: VNN with Read Only Application Intent and database name

Connect to Database Engine

Microsoft SQL Server 2012

Server type: Database Engi

Server name: AGVNN

Authentication: Windows Auth

User name: SOLIDQ\Ad

Password:

Rememb

Connect Cancel

Connect to Database Engine

Microsoft SQL Server 2012

Login | Connection Properties | Additional Connection Parameters

Type or select the name of the database for the connection.

Connect to database: AGDB01

Network

Network protocol: <default>

Network packet size: 4096 bytes

Connection

Connection time-out: 15 seconds

Execution time-out: 0 seconds

Encrypt connection

Use custom color: Select...

Reset All

Connect Cancel Help Options <<

Connect to Database Engine

Microsoft SQL Server 2012

Connection Properties | Additional Connection Parameters

additional connection string parameters (will be sent in clear text):

ationIntent=ReadOnly

Connection string parameters override graphical selections on panels)

Connect Cancel Help Options <<

Microsoft SQL Server

Step-by-step AG

- Accessing AG: Read Only Routing working!

simple-queries.sql - AGVNN.AGDB01 (SOLIDQ\Administrator (52)) - Microsoft SQL Server Management Studio (Administrator)

File Edit View Query Project Debug Tools Window Help

New Query | AGDB01 | Execute | Debug

Object Explorer

Connect

- SQL1 (SQL Server 11.0.2100 - SOLIDQ\Ac
  - Databases
    - System Databases
    - Database Snapshots
    - AGDB01
    - AGDB02
  - Security
  - Server Objects
  - Replication
  - AlwaysOn High Availability
    - Availability Groups
      - AlwaysOnAG01 (Primary)
        - Availability Replicas
        - Availability Databases
        - Availability Group Listener
  - Management
  - Integration Services Catalogs
  - SQL Server Agent

simple-queries.sql -...DQ\Administrator (52)) | AlwaysOnAG01:SQL2 | AlwaysOnAG01:SQL1 | Dashboard: SQL1

```
SELECT  @@SERVERNAME AS Instance;
GO
SELECT * FROM AGDB01.dbo.T1;
SELECT * FROM AGDB02.dbo.T2;
GO
```

100 %

Results

```
Instance
--------------------------------------------------------------------------------
SQL2

(1 row(s) affected)

F1
-----------
1
2
3
4
5

(5 row(s) affected)

F1
-----------
1
2
3
4
5

(5 row(s) affected)
```

100 %

Query executed successfully. | AGVNN (11.0 RTM) | SOLIDQ\Administrator (52) | AGDB01 | 00:00:00 | 11 rows

Ready | Ln 4 | Col 29 | Ch 29 | INS

Microsoft SQL Server

AlwaysOn AG

▶ Read-Only Access Capacity Planning

- Synchronization latency of replica can raise because of the read I/O activity
- Impact on tempdb
  - Temporary statistics
    - Automatically created
    - Gone after failover
  - Read Committed Snapshot Isolation
    - 14 bytes added to rows at primary (on updates)

Microsoft SQL Server

AlwaysOn AG

▶ Problem with replicas at database level

- Need Manual synchronization for objects outside database scope
  - New logins/users
  - Jobs
- Solution for logins/users
  - Contained databases
  - Problem with ApplicationIntent =ReadOnly

Microsoft SQL Server

Advanced AlwaysOn Scenarios

▶ WSFC: Quorum, votes and DR site

- Quorum
  - Majority Node, Majority Node and Fileshare
  - Can’t use Quorum Disk without shared storage
- Windows 2008 e 2008 R2 + QFE
  - Allow to assign voting rights to nodes
  - Avoid scenarios where DR site takes over
  - Common scenario is to always failover manually to DR site i.e. DR site is not considered for HA

Microsoft SQL Server

Advanced AlwaysOn Scenarios

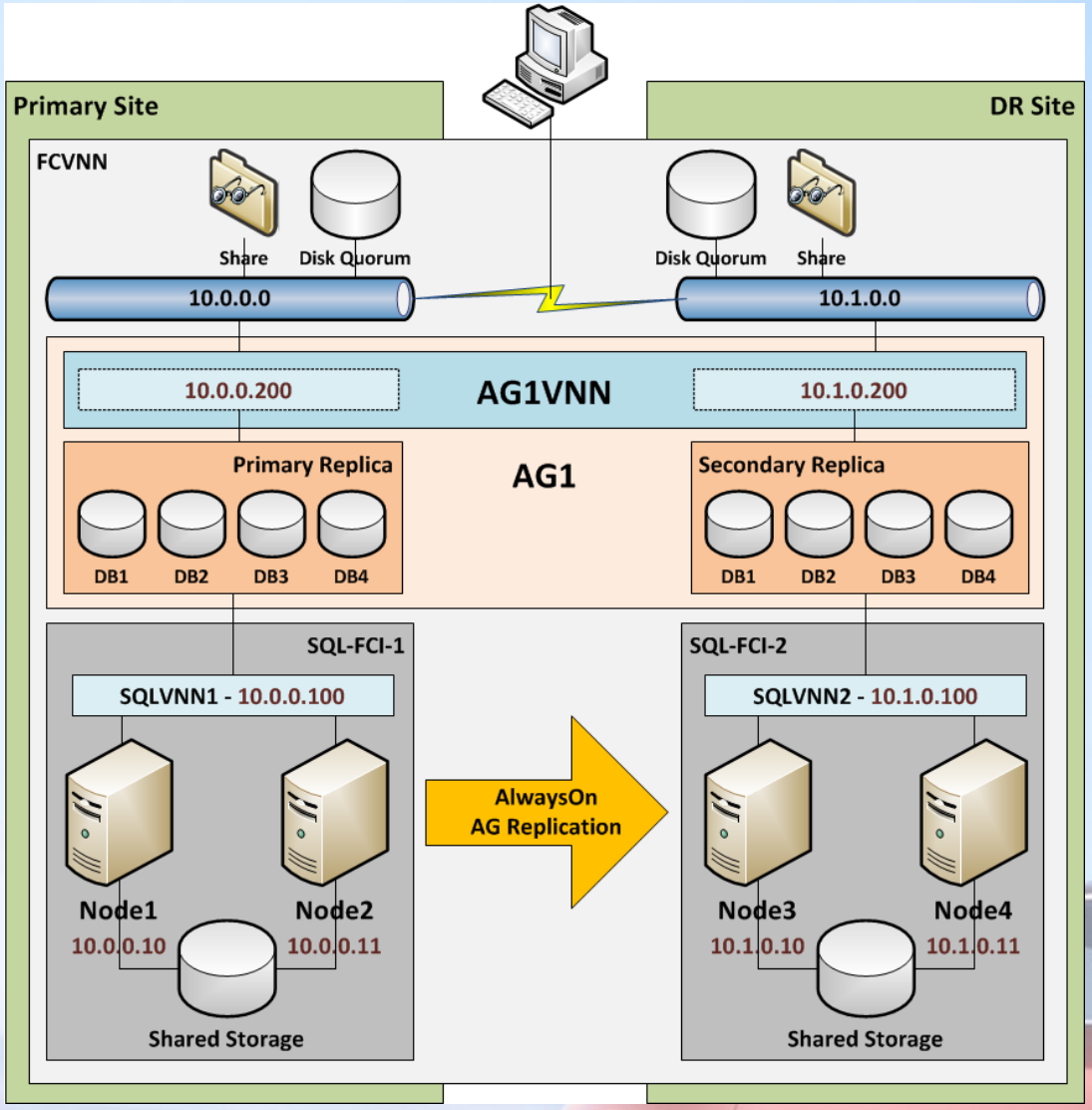
▶ WSFC: Asymmetric Storage

- Windows 2008 + QFE or 2008 R2 + SP1
- Disks shared only among some nodes
  - E.g. one SAN on primary site and one SAN on DR site, volumes shared only among local nodes while the cluster streches across sites
- Allows new Quorum models
  - Asymmetric Disk Majority
  - Node Majority + Asymmetric Disk Majority

Microsoft SQL Server

Advanced AlwaysOn Scenarios

- Multi-site FCI+AG scenario



Microsoft SQL Server

Advanced AlwaysOn Scenarios

- Topology Example 2

Shared storage, regional and geo secondaries

Asynchronous data movement

Microsoft SQL Server

HA/DR Editions and Support

▶ Minimum Support for non Enterprise editions

- Server Core
- Log Shipping (except on Express)
- Database Mirroring
  - Deprecated!
  - Synchronous only for BI/Standard
  - Witness only for Web/Express
- Failover Clustering Instance
  - 2 nodes for BI/Standard
- Purely passive nodes licensing as in 2008

Microsoft SQL Server

AlwaysOn

▶ AG Resources

- Books Online
- SQL AlwaysOn Blog
  - http://blogs.msdn.com/b/sqlalwayson
- CSS SQL Server Engineers
  - http://blogs.msdn.com/b/psssql/archive/tags/alwayson
- SQL CAT
  - http://sqlcat.com

AlwaysOn

▶ AG Whitepapers series: AlwaysOn HA/DR architectures and design patterns

- Multisite Failover Cluster Instance
  - http://sqlcat.com/sqlcat/b/whitepapers/archive/2011/12/22/sql-server-2012-alwayson_3a00_-multisite-failover-cluster-instance.aspx
- SQL Server 2012 AlwaysOn High Availability and Disaster Recovery Design Patterns
  - http://sqlcat.com/sqlcat/b/msdnmirror/archive/2011/12/22/sql-server-2012-alwayson-high-availability-and-disaster-recovery-design-patterns.aspx
- Building a High Availability and Disaster Recovery Solution by Using AlwaysOn Availability Groups
  - http://sqlcat.com/sqlcat/b/whitepapers/archive/2012/06/22/alwayson-architecture-guide-building-a-high-availability-and-disaster-recovery-solution-by-using-alwayson-availability-groups.aspx
- Building a High Availability and Disaster Recovery Solution by Using Failover Cluster Instances and Availability Groups
  - http://sqlcat.com/sqlcat/b/whitepapers/archive/2012/06/22/alwayson-architecture-guide-building-a-high-availability-and-disaster-recovery-solution-by-using-failover-cluster-instances-and-availability-groups.aspx

Microsoft SQL Server

AlwaysOn

▶ AG Resources: articles

- Dashboard
  - http://www.sqlskills.com/blogs/joe/post/Answering-Questions-with-the-AlwaysOn-Dashboard.aspx
- Health Model
  - http://blogs.msdn.com/b/sqlalwayson/archive/2012/02/09/overview-of-the-alwayson-manageability-health-model.aspx
  - http://blogs.msdn.com/b/sqlalwayson/archive/2012/02/13/extending-the-alwayson-health-model.aspx
- Client Reconnect
  - http://www.mattmasson.com/index.php/2012/03/alwayson-multi-subnet-failover-and-ssis

Microsoft SQL Server

Questions & Answers

Microsoft SQL Server

Grazie!

Non dimenticate di compilare i moduli di feedback.

Commenta la sessione che hai appena seguito su Twitter #sqlstart

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